Data Processing & Transformation Log

Project

PRODIOSLABS | Data Analyst Assignment

Objective

Analyze candidate behavior during an online examination using event-level telemetry logs and develop insights through SQL analysis and Power BI visualization.

Dataset Overview

- Database: exam_event_logs
- Table: candidate_log
- Approximate Candidates: 88,000+
- Event Records: 8.2M+
- Examination Window: 09:00 AM – 07:15 PM

Data Import Process

1. Restored the PostgreSQL dump file into the PostgreSQL database.
2. Verified successful restoration of the candidate_log table.
3. Connected PostgreSQL with pgAdmin 4 for data exploration and querying.

Data Validation

The following validation checks were performed:

- Total row count verification.
- Distinct candidate count verification.
- Activity distribution analysis.
- Language distribution verification.
- Question section validation.
- Question response validation.
- Missing value inspection.

Data Quality Observations

- Approximately 90.73% of records were Auto Save events.
- Auto Save events represent system-generated logs rather than direct candidate actions.
- Login, exam start, and navigation events were not available in the dataset.
- No answer keys or item identifiers were provided, preventing score calculation.

Data Transformation Steps

Activity Analysis

Grouped event records by activity type to identify dominant user and system actions.

Candidate Segmentation

Candidates were classified into:

- Light Engagement (<20 events)
- Medium Engagement (20–49 events)
- Heavy Engagement (50+ events)

Section Analysis

Aggregated interaction counts by question section to understand section-wise engagement.

Language Analysis

Grouped records by question language to identify language preferences.

Response Analysis

Analyzed answer option selections (A, B, C, D) to identify response patterns.

Time-Based Analysis

Aggregated events by hour to identify peak activity periods during the examination.

SQL Queries Used

- Activity Analysis
- Candidate Segmentation
- Section Analysis
- Language Analysis
- Response Analysis
- Hourly Activity Analysis

Power BI Transformation

The exported SQL outputs were imported into Power BI and visualized using:

- KPI Cards
- Bar Charts
- Donut Charts

- Line Charts
- Slicers

Assumptions

- Auto Save events were treated as system-generated interactions.
- Candidate engagement was measured using event frequency.
- Response selections were analyzed without evaluating correctness.